Enhancing Image Classification Performance Using Conditional Generative Data Augmentation

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Abstract

Deep learning image classifiers often require large labeled datasets to achieve strong performance. However, collecting labeled data can be expensive, time-consuming, or limited by privacy and access constraints. This project investigates whether synthetic images generated by conditional generative models can improve downstream image classification performance, and under what conditions this improvement is most meaningful. Experiments were conducted on MNIST and Fashion-MNIST using a Conditional GAN (CGAN), with additional comparison experiments using a Conditional VAE-GAN hybrid.

The CGAN experiments show that synthetic data is most useful when real labeled data is scarce. On Fashion-MNIST with only 100 real samples, accuracy improved from 63.6% to 70.8% (+7.1 pp). This benefit decreased as more real data became available and nearly disappeared around 6,000 real samples. Architecture

ablations showed all variants fall within a narrow 91-92% accuracy band, while training strategy especially the learning-rate ratio was the strongest performance lever.

The VAE-GAN experiments provide an important contrast. Across both datasets, VAE-GAN synthetic augmentation generally failed to outperform real-only training. The only consistently positive configuration was large-latent-256 (+0.07 pp MNIST, +0.14 pp Fashion-MNIST). Increasing the synthetic mix ratio degraded Fashion-MNIST performance monotonically. Overall, synthetic data can be valuable as a supplement under scarcity, but its effectiveness depends strongly on generator quality, dataset complexity, and mixing strategy. Synthetic data should not be treated as a replacement for real labeled data.

1 Introduction

Deep learning models have achieved strong performance in image classification tasks, but they typically depend on large labeled datasets. In many practical domains, labeled data can be expensive to collect, difficult to annotate, or limited by privacy constraints. This motivates the use of generative models to create synthetic training examples that supplement real datasets.

Generative Adversarial Networks (GANs) provide one approach for generating synthetic images. A GAN consists of two competing neural networks: a generator that produces synthetic samples and a discriminator that attempts to distinguish real images from generated images. Through adversarial training, the generator learns to produce samples that resemble the real data distribution.

This project investigates whether GAN-generated synthetic images can improve image classification performance. The final study extends the interim report by using a Conditional GAN (CGAN), adding Fashion-MNIST as a more difficult dataset, evaluating data scarcity, performing architecture and training-strategy ablations, comparing multiple ways of using synthetic data in classifier training, and incorporating a Conditional VAE-GAN hybrid study as an additional generative-model comparison. The full CGAN pipeline ran for 198.9 minutes on Apple Silicon MPS.

2 Problem Statement

Traditional data augmentation techniques such as flipping, rotation, scaling, and cropping can increase training-set diversity, but they are limited because they transform existing images rather than generating genuinely new examples. GANs can generate novel images that attempt to follow the real data distribution, making them a potential tool for improving classifier generalization.

However, synthetic data is not automatically useful. Poorly generated images may introduce noise, class ambiguity, or distribution mismatch. It is also unclear whether synthetic data is useful in full-data settings, low-data settings, or both. A further question is whether more complex generative approaches such as VAE-GAN hybrids produce samples that are more useful for augmentation than a standard CGAN. The objective of this project is to evaluate whether conditional generative models can improve image classification performance and to identify the conditions under which synthetic augmentation is most beneficial.

3 Research Questions

This study is guided by the following research questions:

- **RQ1:** Can cGAN-generated synthetic images improve downstream classifier performance?
- **RQ2:** How does classifier performance vary across real-only, real plus synthetic, synthetic-only, weighted mixing, synthetic pretraining, and progressive training?
- **RQ3:** Does synthetic augmentation help more when real labeled data is scarce?
- **RQ4:** How do GAN architecture choices (generator type, latent dimension, embedding dimension, hidden depth, discriminator dropout) affect downstream classifier performance?

- **RQ5:** How do GAN training strategies such as learning-rate ratio, loss function, label smoothing, and training epochs affect augmentation quality?
- **RQ6:** How does the effect of synthetic augmentation differ between MNIST and Fashion-MNIST?
- **RQ7:** Does a Conditional VAE-GAN hybrid provide more useful synthetic data than the cGAN approach?
- **RQ8:** What is the best practical strategy for using synthetic data in classifier training?

4 Related Work and Background

Generative Adversarial Networks were introduced by Goodfellow et al. [1], establishing an adversarial framework in which a generator and discriminator are trained simultaneously. The generator learns to create realistic samples while the discriminator learns to distinguish generated from real data.

Conditional GANs were proposed by Mirza and Osindero [2]. This approach conditions both the generator and discriminator on class labels, allowing class-specific image generation. This is particularly useful for supervised image classification tasks, where generated images must correspond to known labels.

Variational Autoencoders (VAEs) were introduced by Kingma and Welling [3]. VAEs learn a regularized latent representation of data, enabling smoother sampling and reconstruction. VAE-GAN hybrid models combine the structured latent space of VAEs with adversarial image-quality objectives from GANs [4].

Data Augmentation GANs explored the direct use of GAN-generated examples for improving classifier performance, especially in low-data settings [5]. More advanced architectures, including StyleGAN, further improved image quality and realism, motivating the use of generative models as augmentation tools [6].

5 Methodology

5.1 Datasets

- **MNIST:** 60,000 training and 10,000 test images of handwritten digits (0-9). Grayscale, 28×28 pixels. Used as the simpler baseline; real-only performance exceeds 99%, leaving minimal augmentation headroom.
- **Fashion-MNIST:** 60,000 training and 10,000 test images of clothing items in the same 28×28 grayscale format. Visually more complex with overlapping classes, providing more meaningful room for augmentation effects.

5.2 Conditional GAN Architecture

The main project used a Conditional GAN (CGAN). Both generator and discriminator receive class label information. The default configuration was:

- Generator input: random noise $z \in R^{100}$ concatenated with class label y
- Generator: MLP, hidden dimensions $256 \rightarrow 512 \rightarrow 1024$, output: 28×28 grayscale image
- Discriminator: MLP, hidden dimensions $512 \rightarrow 256 \rightarrow 1$
- Conditioning: learned class embedding, default dimension 50
- Optimizer: Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$, $LR = 2 \times 10^{-4}$)

5.3 Conditional VAE-GAN Hybrid Extension

A Conditional VAE-GAN was implemented as an extension to evaluate whether a hybrid latent-variable model could produce more useful synthetic images. The model combines a Variational Autoencoder with a GAN-style discriminator and contains three components:

- Encoder: Maps (image, class label) $\rightarrow$ latent parameters and $\log \sigma^2$

- Generator/Decoder: Samples z from latent space → generates class-conditioned images
- Discriminator: Distinguishes real from generated images, also receives class labels

The training objective combines reconstruction loss, KL-divergence, and adversarial loss:

$$\begin{aligned}
L_D &= BCE(D(x_{real}, y), 1) + \frac{1}{2}[BCE(D(x_{rec}, y), 0) \\
&\quad + BCE(D(x_{gen}, y), 0)] \\
L_{E+G} &= w_{rec} \cdot MSE(\hat{x}, x) + w_{KL} \cdot KL(q||p) \\
&\quad + w_{adv} \cdot \frac{1}{2}[L_{adv,rec} + L_{adv,gen}]
\end{aligned}$$

Default loss weights: $w_{rec} = 1.0$, $w_{KL} = 0.001$, $w_{adv} = 0.1$.

5.4 Classifier Architecture

All scenarios used the same CNN classifier architecture to isolate the effect of synthetic data from classifier capacity:

- Convolution (32 filters) → Max Pooling
- Convolution (64 filters) → Max Pooling
- Fully Connected (128 hidden units) → Fully Connected (10 output classes)
- Optimizer: Adam ($lr = 1 \times 10^{-3}$) cross-entropy loss. Evaluated on the official 10,000-image test split.

6 Experimental Setup

The experimental study consisted of twelve components:

1. Main CGAN scenario comparison across MNIST and Fashion-MNIST

2. CGAN architecture ablation (generator type, conditioning, latent dim, embedding dim, hidden depth, discriminator dropout)
3. CGAN training strategy ablation (loss type, LR ratio, label smoothing, GAN training epochs)
4. CGAN data scarcity analysis (100 to 12,000 real samples on Fashion-MNIST)
5. CGAN synthetic volume sweep (500 to 12,000 synthetic images on Fashion-MNIST)
6. CGAN conditioning and recognizability study (embedding, projection, one-hot)
7. VAE-GAN three-way classifier comparison (real-only, real+synthetic, synthetic-only)
8. VAE-GAN architecture variant ablation (7 configurations across two datasets)
9. VAE-GAN training strategy sweep (9 loss/LR/batch-size configurations)
10. VAE-GAN data scarcity analysis (500 to 60,000 real samples on both datasets)
11. VAE-GAN label injection method comparison (3 conditioning methods)
12. VAE-GAN extended augmentation scenarios (8 mix strategies on both datasets)

The full CGAN pipeline ran for 198.9 minutes on Apple Silicon MPS. All classifiers were evaluated on the official 10,000-image test split.

7 CGAN Results and Analysis

7.1 Main CGAN Scenario Comparison

The MNIST results are near ceiling across all real-data strategies ($\geq$99.06%), leaving little room for augmentation to improve performance. Fashion-MNIST is more informative with a real-only baseline of 92.29%. The weighted real-plus-synthetic

Table 1: Top-Line CGAN Scenario Comparison

Scenario	MNIST Acc.	MNIST F1	F-MNIST Acc.	F-MNIST F1
real_only	99.27%	99.27%	92.29%	92.25%
augmented_real	99.13%	99.13%	87.73%	87.45%
real_plus_synthetic	99.13%	99.13%	91.66%	91.74%
synthetic_only	66.78%	66.32%	66.54%	66.21%
real_plus_synthetic_weighted	99.16%	99.16%	92.27%	92.18%
synthetic_pretrain	99.06%	99.06%	91.03%	91.04%
progressive	99.14%	99.14%	91.72%	91.78%

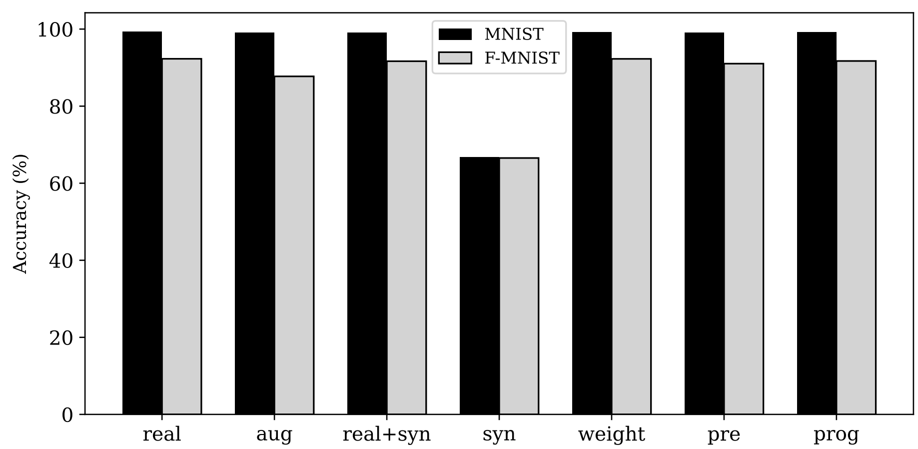

Figure 1: Main scenario comparison across MNIST and Fashion-MNIST. MNIST is near ceiling; Fashion-MNIST reveals meaningful strategy differences.

strategy nearly matched this baseline at 92.27% (only -0.02 pp below). Synthetic-only training collapsed to 66.54%-66.78% across both datasets, confirming that generated images cannot replace real labeled data. Geometric augmentation performed especially poorly on Fashion-MNIST (87.73%), likely because clothing items are orientation-sensitive and simple geometric transforms create semantically invalid examples.

7.2 CGAN Architecture Ablation

Table 2: Generator Type Ablation

Generator Type	Accuracy	F1 Score	Precision	Recall
MLP	0.9178	0.9177	0.9180	0.9178
DCGAN	0.9129	0.9133	0.9143	0.9129

The MLP generator outperformed DCGAN by 0.49 pp for 28×28 grayscale images. Simpler MLP architecture achieved better downstream accuracy, suggesting that convolutional generators do not provide additional benefit for small grayscale images.

Table 3: Conditioning Method Ablation

Conditioning Method	Accuracy	F1 Score	Precision	Recall
Embedding, learned dim=50	0.9132	0.9130	0.9137	0.9132
One-hot, fixed dim=10	0.9118	0.9118	0.9129	0.9118
Projection discriminator	0.9143	0.9144	0.9150	0.9143

Projection conditioning produced the highest downstream accuracy (91.43%). One-hot conditioning performed worst (91.18%). Richer label conditioning improves class-specific generation quality; sparse one-hot vectors are less effective for preserving class identity.

Table 4: Architecture Ablation Summary

Factor	Best Config	Best Acc.	Worst Config	Worst Acc.	Gap
Generator type	MLP	91.78%	DCGAN	91.29%	0.49 pp
Latent dimension	$z = 100$	91.79%	$z = 200$	91.31%	0.48 pp
Embedding dim.	emb=32	91.54%	$emb = 10$	91.05%	0.49 pp
Hidden depth	256x512	91.71%	256x512x1024	91.57%	0.14 pp
Disc. dropout	0.0	91.75%	0.3	91.35%	0.40 pp

Table 5: LR Ratio Ablation

LR Ratio	Accuracy	F1 Score	Precision	Recall
G=4e-4, D=2e-4 (faster gen)	0.9204	0.9201	0.9203	0.9204
G=2e-4, D=2e-4 (equal rates)	0.9143	0.9142	0.9148	0.9143
G=2e-4, D=4e-4 (faster disc)	0.9131	0.9129	0.9137	0.9131
G=1e-4, D=4e-4 (much faster)	0.9128	0.9127	0.9139	0.9128

7.3 CGAN Training Strategy Ablation

LR ratio was the most impactful training factor. A faster generator LR ($G = 4 \times 10^{-4}, D = 2 \times 10^{-4}$) achieved 92.04% accuracy the highest across all cGAN experiments. Giving the generator more update momentum relative to the discriminator improves sample diversity and augmentation quality.

Table 6: Training Strategy Summary

Factor	Best Config	Best Acc.	Worst Config	Worst Acc.	Gap
LR ratio (G:D)	4e-4: 2e-4	92.04%	1e-4: 4e-4	91.28%	0.76 pp
Epochs	ep=15	91.70%	ep=30	91.04%	0.66 pp
Loss function	BCE	91.53%	WGANGP	91.27%	0.26 pp
Label smoothing	None	91.26%	0.9	91.21%	0.05 pp

7.4 CGAN Data Scarcity Analysis

The largest improvement occurred with only 100 real samples (+7.1 pp: 63.6%$\rightarrow$70.7%). The gain shrank monotonically as real data increased, nearly disappearing at 6,000 samples. This strongly supports the conclusion that synthetic data is most useful when the classifier lacks enough real examples to form stable decision boundaries.

Table 7: Fashion-MNIST Data Scarcity Results

Real Samples	Real Only	Real + Syn	Gain (pp)
100	63.6%	70.7%	+7.1
300	71.5%	74.8%	+3.3
600	76.2%	78.2%	+2.0
1,200	79.7%	81.0%	+1.3
3,000	83.0%	84.0%	+1.1
6,000	86.2%	86.1%	=0
12,000	87.8%	87.9%	+0.1

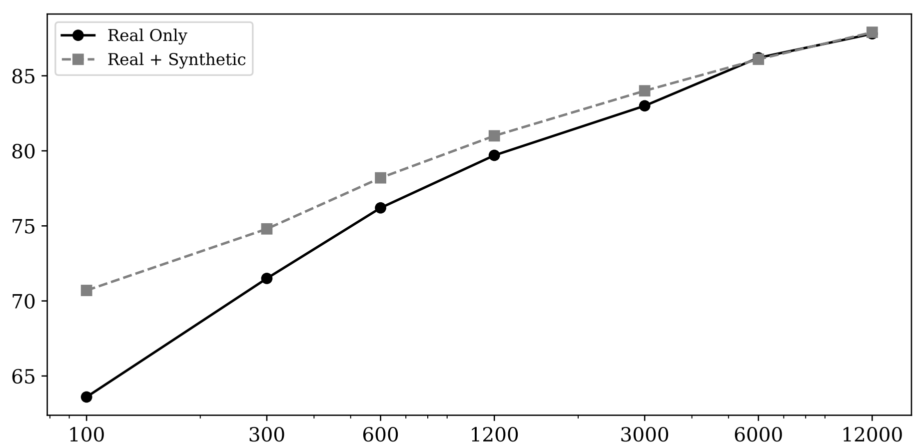

Figure 2: Accuracy vs number of real training samples. The line (real+synthetic) consistently exceeds the (real-only) at low label counts.

Above the crossover point ($\sim$6,000 samples), imperfect synthetic images provide little additional value and may introduce marginal noise.

7.5 CGAN Synthetic Volume Sweep

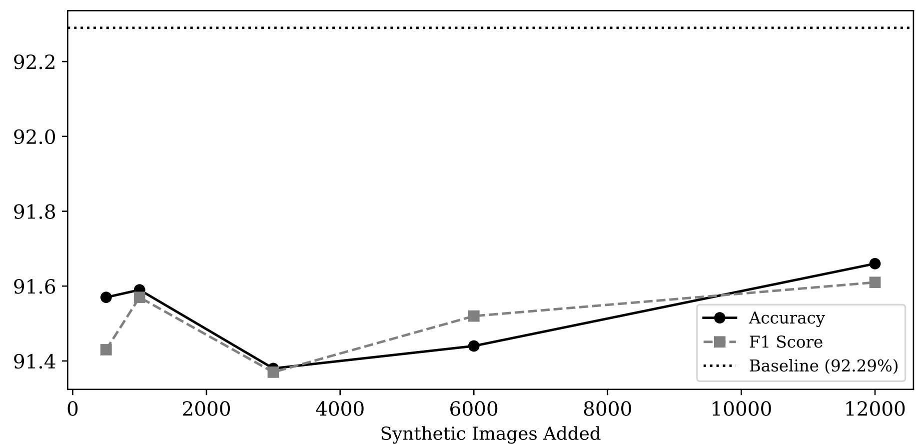

Figure 3: Accuracy and F1 score vs number of synthetic images. No volume reaches baseline.

Adding more synthetic images did not consistently improve performance in the full-data setting. Even with 12,000 synthetic images, accuracy reached only 91.66% - still below the 92.29% real-only baseline. This confirms that synthetic volume alone does not guarantee improvement when real data is already abundant.

7.6 CGAN Conditioning and Recognizability

Projection conditioning achieved the highest downstream accuracy (91.43%), while learned embedding achieved the highest recognizability (99.9%). One-hot conditioning performed worst on both metrics. Importantly, highest recognizability does not correspond to highest downstream accuracy: learned embeddings produce the most class-consistent images, but projection conditioning produces images that improve classifier performance more effectively.

8 Conditional VAE-GAN Hybrid Study

The Conditional VAE-GAN experiments were added as an extension to evaluate whether a hybrid latent-variable generative model could produce more useful synthetic images than the CGAN. The motivation was that a VAE-GAN may provide smoother latent sampling through the VAE component while still benefiting from adversarial training through the GAN discriminator.

8.1 VAE-GAN Goal 2: Three-Way Classifier Comparison

Table 8: Goal 2 Results - MNIST

Scenario	Accuracy	Precision	Recall	F1
real_only	0.9925	0.9925	0.9925	0.9925
real_plus_syn	0.9910	0.9910	0.9910	0.9910
synthetic_only	0.9336	0.9361	0.9336	0.9340

Table 9: Goal 2 Results - Fashion-MNIST

Scenario	Accuracy	Precision	Recall	F1
real_only	0.9195	0.9202	0.9195	0.9196
real_plus_syn	0.9175	0.9177	0.9175	0.9173
synthetic_only	0.7913	0.7971	0.7913	0.7918

Unlike the CGAN scarcity results, the VAE-GAN three-way comparison showed that adding synthetic data slightly reduced performance on both datasets. On MNIST, F1 decreased from 0.9925 to 0.9910. On Fashion-MNIST, F1 decreased from 0.9196 to 0.9173. Synthetic-only VAE-GAN training performed better than earlier CGAN synthetic-only runs (especially on MNIST: 93.36% vs 66.78%), but still remained far below real-only training. This suggests that VAE-GAN samples contained useful class information, but not enough diversity or distributional accuracy to improve the classifier when real data was already available.

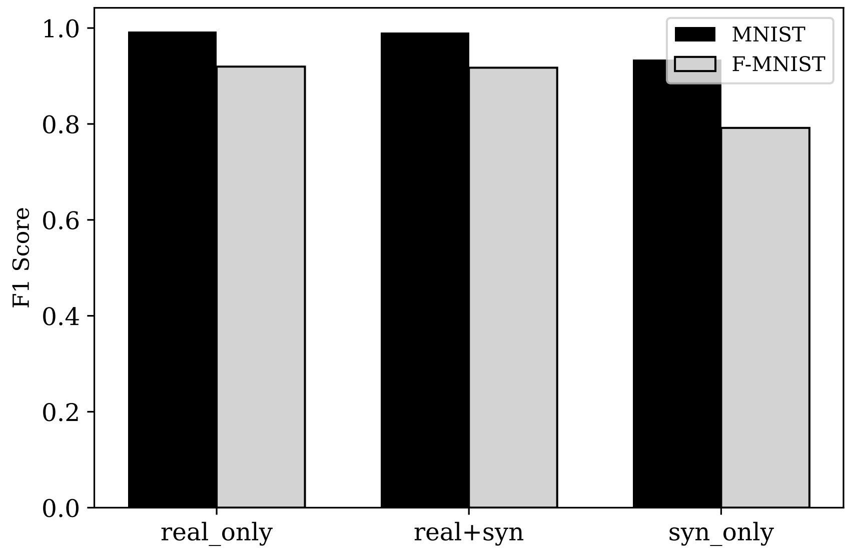

Figure 4: Goal 2 F1 Scores for VAE-GAN across datasets.

8.2 VAE-GAN Goal 3: Architecture Variants

Table 10: Goal 3 Results: F-MNIST ΔF1 (pp)

Variant	Real F1	Real+Syn	ΔF1
base_mlp_shallow	0.9196	0.9173	-0.24
large_latent_256	0.9196	0.9210	+0.14
small_latent_32	0.9196	0.9186	-0.10
deep_mlp	0.9196	0.9198	+0.02
cnn_generator	0.9196	0.9134	-0.62
no_disc_dropout	0.9196	0.9186	-0.10
high_disc_dropout	0.9196	0.9145	-0.51

The most important finding is that *large_latent_256* was the only VAE-GAN configuration that improved performance on both datasets (+0.07 pp MNIST, +0.14 pp Fashion-MNIST). A larger latent space helped the VAE-GAN capture more useful image variation. The CNN generator performed especially poorly on Fashion-MNIST (-0.62 pp), suggesting that transposed-convolution artifacts or poor texture reconstruction confused the downstream classifier. High discriminator dropout consistently reduced performance on both datasets.

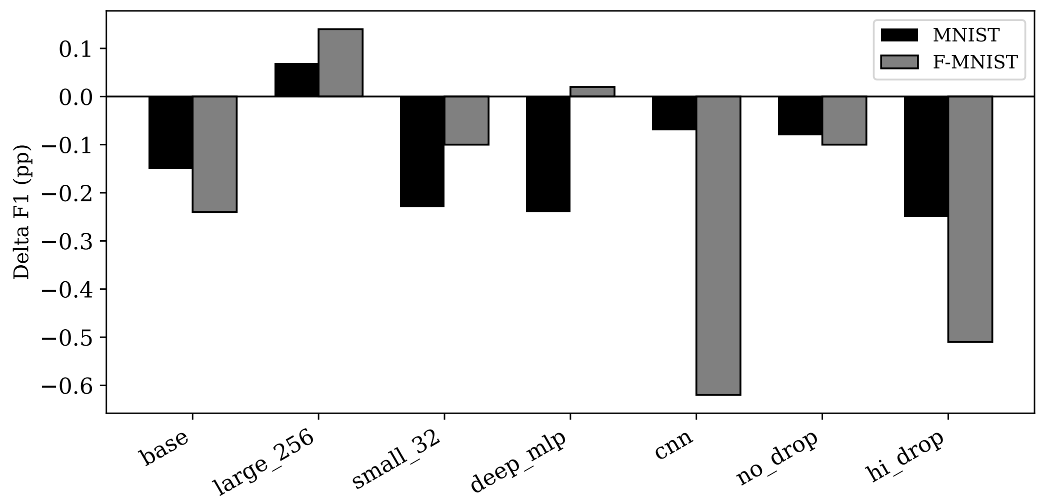

Figure 5: Goal 3 Delta F1 scores for various VAE-GAN architecture variants.

8.3 VAE-GAN Goal 4: Training Strategies

Table 11: Goal 4 Results: F-MNIST ΔF1 (pp)

Strategy	Real F1	Real+Syn	ΔF1
balanced	0.9196	0.9173	-0.24
recon_heavy	0.9196	0.9161	-0.35
adv_heavy	0.9196	0.9183	-0.13
kl_heavy	0.9196	0.9177	-0.19
no_kl	0.9196	0.9180	-0.16
low_lr	0.9196	0.9163	-0.33
high_lr	0.9196	0.9178	-0.18
large_batch	0.9196	0.9179	-0.17
small_batch	0.9196	0.9140	-0.56

Most VAE-GAN training modifications did not improve downstream performance. Small batch size produced a very small positive gain on MNIST (+0.01 pp) but was the worst strategy on Fashion-MNIST (-0.56 pp). Low learning rate was harmful on both datasets, suggesting insufficient generator convergence. Adversarial-heavy training was the least harmful Fashion-MNIST configuration (-0.13 pp), suggesting that image fidelity matters more than reconstruction smoothness for clothing augmentation.

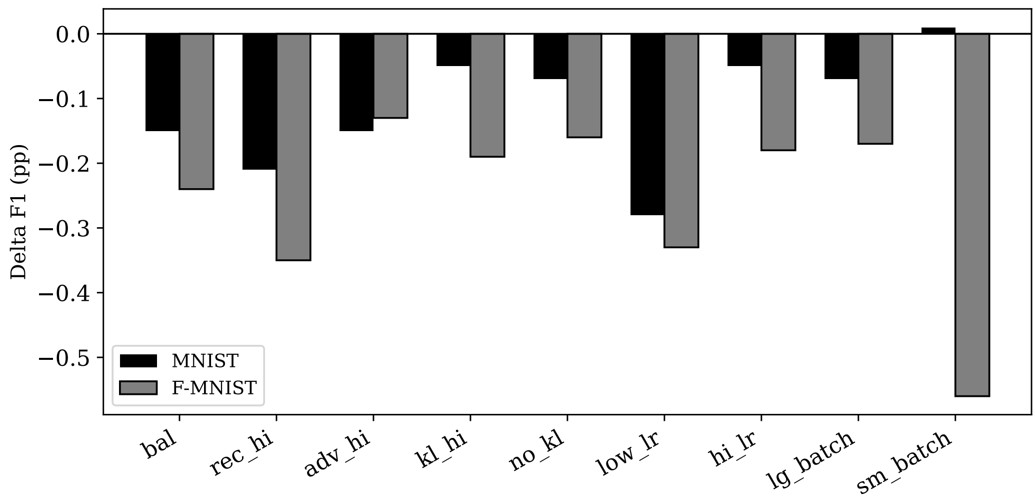

Figure 6: Impact of different VAE-GAN training strategies on F1 score relative to baseline.

8.4 VAE-GAN Goal 5: Data Scarcity Results

Table 12: Goal 5 Scarcity Results - Fashion-MNIST

Setting	Real N	Real-Only F1	Real+Syn F1	ΔF1
full_real	60,000	0.9196	0.9173	-0.24 pp
5k_real	5,000	0.8551	0.8402	-1.49 pp
2k_real	2,000	0.8141	0.8102	-0.39 pp
500_real	500	0.7526	0.7515	-0.11 pp

The VAE-GAN scarcity results contrast sharply with CGAN findings. Instead of helping under scarcity, VAE-GAN synthetic augmentation reduced F1 at all tested levels. The largest penalties occurred under moderate scarcity: on MNIST with 2,000 real examples, F1 dropped by -2.49 pp; on Fashion-MNIST with 5,000 real examples, by -1.49 pp. This reveals a double weakness: scarcity degrades both generator quality (fewer real examples to learn from) and classifier anchoring (fewer real examples to resist synthetic noise). The very low scarcity setting (500 real) showed smaller penalties, likely because the classifier had little real data to disrupt in the first place.

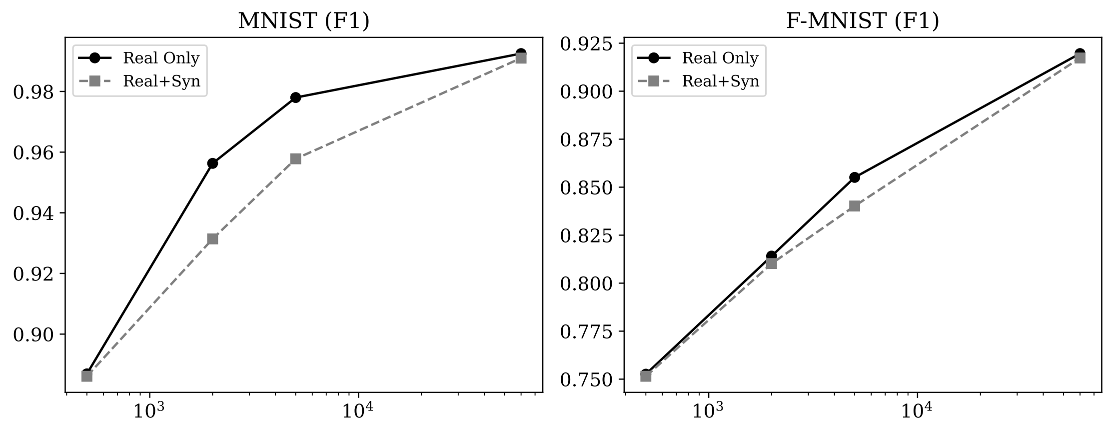

Figure 7: F1 scores under data scarcity for both MNIST and Fashion-MNIST using VAE-GAN.

8.5 VAE-GAN Goal 6: Label Injection Results

Table 13: Goal 6 Label Injection Methods - ΔF1

Method	MNIST ΔF1	F-MNIST ΔF1
learned_embedding	-0.15 pp	-0.24 pp
onehot_labels	-0.37 pp	-0.30 pp
gen_only_embedding	-0.14 pp	-0.28 pp

One-hot label injection performed worst overall (-0.37 pp on MNIST). Learned embeddings were more stable. Generator-only embedding produced the smallest MNIST penalty (-0.14 pp). Learned label representations are preferable to sparse one-hot vectors for class-conditioned VAE-GAN generation, consistent with the cGAN conditioning ablation findings.

8.6 VAE-GAN Goal 7: Extended Augmentation Scenarios

The extended augmentation results confirm that increasing VAE-GAN synthetic data does not improve performance. On Fashion-MNIST, F1 decreased monotonically from 25% to 200% mix ratio. Weighted sampling (-0.48 pp) reduced the damage more than equal mixing (-0.70 pp), but still could not recover the real-only baseline. Synthetic pretraining followed by real fine-tuning was the worst strategy

Table 14: Goal 7 Scenarios - Fashion-MNIST

Scenario	Accuracy	F1	ΔF1
real_only	0.9195	0.9196	-
real + 25% syn	0.9190	0.9181	-0.15 pp
real + 50% syn	0.9129	0.9127	-0.70 pp
real + 100% syn	0.9127	0.9127	-0.70 pp
real + 200% syn	0.9087	0.9094	-1.02 pp
weighted mix (real 2x)	0.9149	0.9149	-0.48 pp
progressive aug	0.9106	0.9102	-0.95 pp
syn_pretrain → fine-tune	0.8728	0.8721	-4.75 pp

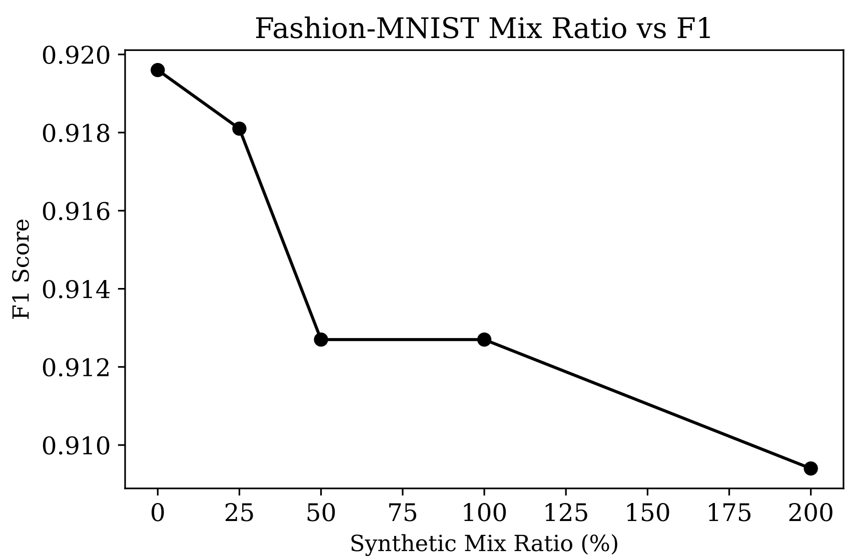

Figure 8: Fashion-MNIST Mix Ratio vs F1 Score. Increasing synthetic proportion monotonically degrades performance.

on Fashion-MNIST (-4.75 pp), suggesting that features learned from low-quality VAE-GAN images transferred poorly even after exposure to real images.

9 Discussion

The final results show that synthetic data augmentation is useful only when generated images provide additional information without introducing too much distribution noise. The CGAN experiments support the value of synthetic data under scarcity: with only 100 real Fashion-MNIST samples, adding synthetic images improved accuracy by 7.1 pp. This suggests that GAN-generated data can compensate for missing real examples when the classifier lacks enough data to learn stable decision boundaries.

However, the VAE-GAN experiments provide an important caution. Although VAE-GAN combines latent-space regularization with adversarial training, its synthetic samples generally did not improve downstream classification. In the basic three-way comparison, VAE-GAN real-plus-synthetic slightly underperformed real-only training on both datasets. The synthetic-only VAE-GAN classifiers performed better than earlier CGAN synthetic-only runs, but they still remained far below real-only training.

The VAE-GAN architecture ablation showed that latent-space capacity matters. The large_latent_256 variant was the only configuration that improved F1 on both datasets. This suggests the default latent space was too constrained to capture useful class variation. However, the improvement was very small, indicating that architecture changes alone are insufficient.

The VAE-GAN scarcity study was especially revealing. Unlike the CGAN experiment, VAE-GAN synthetic augmentation hurt performance across all tested scarcity levels. The largest losses occurred under moderate scarcity, where the generator had too little data to learn high-quality samples, but the classifier still had enough real data that noisy synthetic examples could disrupt learning. This creates a double weakness: scarcity simultaneously harms generator quality and classifier anchoring.

The extended augmentation results also show that more synthetic data can make performance worse. On Fashion-MNIST, increasing the VAE-GAN synthetic mix ratio from 25% to 200% monotonically degraded F1. Synthetic pre-training performed especially poorly (-4.75 pp), suggesting that features learned from low-quality synthetic data transfer badly even after fine-tuning on real images.

Together, the cGAN and VAE-GAN findings lead to a nuanced conclusion: synthetic augmentation can help, but only when generated samples are sufficiently realistic, diverse, class-consistent, and used in the right proportion. A more complex generative model is not automatically better. In this project, CGAN augmentation produced the clearest low-data benefit, while VAE-GAN mainly demonstrated the limits of synthetic data when generator quality is insufficient.

10 Challenges and Limitations

- Both datasets are grayscale and low-resolution (28x28). Findings may not generalize directly to natural color datasets such as CIFAR or ImageNet.
- Most ablation runs used a single random seed. More reliable conclusions would require repeated trials with multiple seeds and confidence intervals.
- FID and other image-quality scores are difficult to compare across datasets and should be used as relative metrics rather than absolute truth.
- The MLP generator limits image fidelity. Stronger architectures (deeper DCGAN, StyleGAN, diffusion models) may produce different results.
- The VAE-GAN hybrid improved latent-space structure but did not consistently improve downstream classification.
- The classifier architecture was fixed as a shallow CNN. Stronger classifiers such as ResNet may reduce or change the observed real-versus-synthetic gap.
- Synthetic data can introduce noise when real data is already abundant, suggesting the benefit window is narrow.

- Classical geometric augmentation performed poorly on Fashion-MNIST (-4.56 pp), confirming that augmentation must be dataset-aware.
- Increasing the synthetic-to-real ratio can harm performance, especially when the generator does not fully capture the real data distribution.

11 Future Work

- Repeat each ablation with multiple random seeds and report mean performance with standard deviation and confidence intervals.
- Test the method on more complex color datasets such as CIFAR-10, CIFAR-100, or domain-specific image datasets.
- Compare CGAN and VAE-GAN augmentation against diffusion-based synthetic data generation.
- Evaluate stronger classifier architectures such as ResNet or Vision Transformers to assess the real-vs-synthetic gap under higher-capacity models.
- Perform deeper per-class analysis to identify which classes benefit most or least from synthetic augmentation.
- Explore adaptive synthetic sampling, where difficult or underrepresented classes receive more generated examples.
- Combine synthetic augmentation with carefully selected traditional augmentations in a joint strategy.
- Use classifier-guided filtering to remove low-quality or class-confusing synthetic samples before augmentation.
- Explore latent-space quality control for VAE-GANs: latent traversal, sample rejection, and class-conditional diversity scoring.
- Evaluate FID and LPIPS metrics to quantify sample quality changes across architecture ablations.

12 Conclusion

This project evaluated whether conditional generative models can improve image classification by producing synthetic training data. The final study combined cGAN experiments with additional Conditional VAE-GAN experiments on MNIST and Fashion-MNIST, covering twelve experimental components.

The CGAN results show that synthetic data can improve classifier performance when real labeled data is scarce. The strongest gain occurred on Fashion-MNIST with only 100 real samples: adding GAN-generated images improved accuracy from 63.6% to 70.8% (+7.1 pp). Architecture ablations showed that all variants remain within a narrow 91-92% accuracy band, while training strategy - especially the learning-rate ratio - was the strongest performance lever. The weighted real-plus-synthetic strategy was the best practical approach, nearly matching the real-only baseline (-0.02 pp) while allowing synthetic contribution.

The VAE-GAN experiments showed that synthetic augmentation is not guaranteed to help. Across most settings, real-plus-synthetic VAE-GAN training slightly underperformed real-only training. Only the large-latent-256 architecture produced consistent positive gains (+0.07/+0.14 pp). VAE-GAN augmentation also performed poorly under scarcity and became worse as the synthetic mix ratio increased.

The overall conclusion is that synthetic data should be treated as a careful augmentation tool, not a replacement for real labeled data. The value of synthetic augmentation depends on generator quality, dataset complexity, real-data availability, and the mixing ratio. For this project, cGAN-based augmentation was most useful under strong scarcity, while VAE-GAN results mainly showed that hybrid generative models require stronger image quality before they can reliably improve downstream classification. Future work should focus on improving generator fidelity, testing more complex datasets, and applying classifier-guided sample filtering.

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